Series & Sequence

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↳ What is sequence?

A group of numbers/letters/objects/figures are to be arranged in a proper order in such a way that either row/column/diagonal.... wise is called 'sequence'.

↳ What is series?

If a sequence of numbers/letters/figures are in a proper logic b/w each & every consecutive observations then it is 'series'.

Ex:- Sequence of alphabet :- g a c g b i d f h e

[alphabet are in row wise sequence]

Series of above alphabet:- a b c d e f g h i j

[Sequence of alphabet in alphabetical]

↳ Classification of series & sequence. in to

(i) Number Series (ii) Letter series (iii) Mixed series ---

But we mainly consider number series as major one.

Whatever different logics that we are observing in number series, the same we are observing in a letter series by using their positions in all the ways in terms of numbers.

(129) Number series:- It is classified into mainly 7 types based on priority wise.

(i) Prime Series

(ii) Difference Series [+/-]

(iii) Product Series [×/÷]

* (iv) Combination Series [+,− | −,+ | ×,+ | ÷,+ ; +,× | −,× | ×,− | ÷,× ; +,÷ | −,÷ | ×,÷ | ÷,−]
12 combinations

(v) Square Series

(vi) Cube Series

* (vii) Miscelaneous Series.

(i) Prime Series:- In this first you have to know about all prime numbers within 100. [Total 25 prime Nos in 100]

2, 3, 5, 7, 11, 13, 17, 19, 23, 29; 31, 37, 41, 43, 47,
53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

[Within 50 → 15 prime Nos, 51-100 → 10 prime Numbers].

Example:-1:- 7, 11, 13, 17, 19, 23, ____ ?

Ans:- 29 [general prime series].

Ex-2:- 17, 23, 31, 41, 47, 59, ____?

Ans:- The series look like alternate Prime series. So, answer is: 67

Ex-3:- 11, 13, 19, 31, 47, 71, ____ ?

Ans:- b/w 11&13 → no gap of prime nos.
13&19 → gap of one prime (17)
19&31 → missing of 2 prime (23&29) ....

(190) 31847 → missing of 3 prime (37, 41, 43)

47871 → " " 4 " (53, 59, 61, 67) then

Answer is missing of 5 prime number (73, 79, 83, 89, 97) after 71. which is first prime number after 100 i.e., 101.

Ans:- 11, 13, 19, 31, 47, 71, 101

logic is missing prime numbers in natural format.

(ii) Difference Series:- [Add (+) / Sub (−)]

↳ Increamenting / decreamenting the observations in a proper logical way.

Ex:-4:- 17, 20, 23, 26, 29, 32, —?

Ans:- 17, 20, 23, 26, 29, 32, 35

+3 +3 +3 +3 +3 +3

This logic is 'constant difference series'.

Ex:-5:- 1947, 1940, 1926, 1905, 1877, ?

Ans:- 1947, 1940, 1926, 1905, 1877, 1842

−7 −14 −21 −28 7×5

=7×1 =7×2 =7×3 =7×4 =−35

logic → table difference (−) series.

Ex:-6:- 1785, 1792, 1805, 1824, 1853, ?

1785, 1792, 1805, 1824, 1853, 1890

+7 +13 +19 +29 +37

7, (11), 13, (17), 19, (23), 29, (31), 37, 41 ...

Logic → alternate Prime difference series